SeeRise: Visualizing Emulated Sea Level Rise on Coastal Regions

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Abstract

The advent of sea level rise can have devastating consequences on coastal areas all around the world. Low-lying regions—such as Florida, a state particularly susceptible to sea level rise due to its low-lying topography and extensive coastline—are a major focal point when it comes to modeling sea level rise as they are most vulnerable to changes. Using the method described by ”A Semi-Empirical Approach to Projecting Future Sea-Level Rise” (Rahmstorf 2007), which regresses the rate of sea level rise on surface air temperature anomaly, our team coupled this model with emulators from “ClimateBench v1.0: A Benchmark for Data-Driven Climate Projections” (Watson-Parris 2022) to create a predictor capable of simulating sea level rise in any future emission scenario, not just the ones prescribed by SSPs. This impact is then visualized using high-resolution topography data to assess the potential transformation of Florida’s coastal landscape, which can aid policymakers in developing mitigation and adaptation strategies.

Website: `https://zoeludena.github.io/SeeRiseWebsite`
Code: `https://github.com/zoeludena/SeeRise`
App: `https://seerise-floridaapp.streamlit.app/`

1 Introduction

1.1 First Look

Sea level rise is a pressing global issue that comes alongside climate change. Coastal regions are projected to be or have already been impacted by the rising sea level. Beyond the land getting submerged, sea level rise can also lead to coastal erosion, saltwater intrusion, more frequent flooding, and change in coastal ecosystems. In our project, we focus on one of the most vulnerable regions facing sea level rise—Florida. Florida is especially impacted by sea level rise due to its low elevation, porous limestone foundation, and extensive coastline. Understanding the degree of sea level rise and its impact on the Florida coastline corresponding to different choices of human action is crucial for the local population, policymakers, and many other stakeholders.

1.2 Prior Work

Previous research has established a strong correlation between global temperature rise and sea level changes. ”A Semi-Empirical Approach to Projecting Future Sea-Level Rise” (Rahmstorf 2007) introduced a semi-empirical approach to model sea level rise based on observed temperature trends. This method directly links sea level changes to global mean temperature through historical data analysis. The study demonstrated that sea level rise is accelerating in response to increasing global temperatures, suggesting that traditional projections may underestimate future impacts. His approach provides a basis for our study, where we extend this model to evaluate the specific consequences of sea level rise in Florida. By integrating temperature-based projections with coastal visualization techniques, we build upon prior methodologies to assess localized risks and potential landscape transformations.

“ClimateBench v1.0: A Benchmark for Data-Driven Climate Projections” (Watson-Parris 2022) is the paper we were trying to recreate in Fall 2024. This paper is the first benchmarking framework that uses a set of baseline machine learning models on an Earth System Model to emulate the response of different climate variables. This allows people to predict annual mean global distributions of temperature, diurnal temperature ranges, and precipitation given a wide range of emissions and concentrations of carbon dioxide, methane, sulfur dioxide, and black carbon. This allows the baseline models to explore the unexplored. The paper found that the three most accurate baseline models were neural networks, Gaussian processes, and random forests.

1.3 Description of Data

There are two classes of data files that we need for the emulators. After preprocessing, both will be provided to the various models as input for training, validation, and ultimately testing. Both of these data files are binary `.nc` (NetCDF) files, which are essentially multidimensional data structures with “indexes”. In this case, both of the data files are indexed along “lat” (latitude), “lon” (longitude), and “time” (year).

- Climate Model Input Data
  - “CO2” (Carbon Dioxide) (NOAA Climate.gov 2024). One of Earth’s most important greenhouse gases because it absorbs and radiates heat. It is a stable molecule and can remain in the atmosphere for several thousand years.
  - “CH4” (Methane) (NASA Climate Change 2024). Second largest contributor to climate warming after CO_2. Methane is a much more potent greenhouse gas, but has a much shorter half-life of only 8-9 years.
  - “SO2” (Sulfur Dioxide) (NASA Earth Observatory 2017). Sulfur dioxide can react with the atmosphere to form aerosol particles which helps make clouds. It negatively affects air quality (a critical air pollutant) because it mainly comes from burning coal (coal-fired power plants). It can also react with water vapor to form acid rain.
  - “BC” (Black Carbon) (Office of Environmental Health Hazard Assessment OEHHA). Absorbs light and contributes to climate change by releasing heat energy into the atmosphere. It is considered a short-lived pollutant. They can cause greater warming effects than CO_2 even with its short lifespan. Are causing snow, glaciers, and ice to darken and melt.
- Climate Model Output Data
  - “tas” (Surface Air Temperature). Average monthly surface air temperature two meters above the ground. Measured in Kelvin.
- Sea Level Rise Model Input Data
  - Temperature anomaly. Difference between average yearly surface air temperature two meters above the ground and that in the year 1900, measured in Kelvin.
- Sea Level Rise Model Output Data
  - Sea Level Change. Difference between two consecutive global average sea levels, measured in milimeters (mm).

To compare how our emulators produced Sea Level Rise, we used data from ”The Causes of Sea-Level Rise Since 1900” (Thomas Frederikse, et al 2020) and NASA’s Sea Level Projection Tool (NASA 2021), both provided as Excel files.

The first dataset, `global_basin_timeseries.xlsx`, contains historical sea level time series across different ocean basins:

- Observed GMSL [mean]: The globally averaged mean sea level (GMSL) observations over time.
- Baseline Value (GMSL at 1900): The mean sea level in the year 1900 is subtracted to compute anomalies.
- GMSL Anomaly: The deviation of the observed mean sea level from the 1900 baseline, highlighting long-term sea level changes.

The second dataset, `ipcc_ar6_sea_level_projection_global.xlsx`, includes global sea level rise projections based on IPCC AR6 scenarios:

- Scenario: Specifies the climate scenario being analyzed.

- Quantile: Defines the confidence interval levels used for projections:
  - 5th percentile: Lower bound of projections, represents a conservative estimate of sea level rise.
  - 50th percentile: median estimate of projected sea level rise.
  - 95th percentile: Upper bound of projections, representing a high-end estimate.
- Confidence: Level of confidence in the projections, it keeps confidence with “medium” and “high” estimates.
- Years: Represents different years containing projected sea level rise.

These datasets allowed for direct comparison between past observations and future projections.

2 Methods

2.1 Preparing the Data

Our emulators are fitted to historical data and each Shared Socioeconomic Pathways (SSP) (SSP 126, SSP 370, and SSP 585).

These SSPs look at five different ways the world might evolve with different levels of climate mitigation and policy. We only look at four in this paper. The underlying factors: population, technological, and economic growth could lead to future emissions and warming outcomes (Brief 2023).

- SSP 126 ”Taking the Green Road”: There is an emphasis on human well-being, driven by an increasing commitment to achieve development goals. There is lower material growth and lower resource and energy intensity.
- SSP 245 ”Middle of the Road”: Social, economic, and technological trends do not shift much from historical patterns. Environmental systems experience some degradation and the intensity of resource and energy use declines.
- SSP 370 ”A Rocky Road”: policies shift to become increasingly oriented toward national and regional security issues. Countries focus on achieving their personal goals within their regions. Consumption is material-intensive and there is a low priority for addressing environmental concerns.
- SSP 585 ”Taking the Highway”: World places faith in competitive markets, innovation, and participatory societies to produce technological progress to create a sustainable future. Push for economic and social development is coupled with the exploitation of fossil fuel resources.

We validated our models on SSP 245. The emulators take in a combination of greenhouse gas emissions as inputs: normalized carbon dioxide, normalized methane, principal component black carbon, and principal component sulfur dioxide.

Table 1: The Greenhouse Gas Values in 2025 Assuming SSP 245

Component	Value
CH_4	0.474023
pseudo_pcs, BC_0	1.747441
pseudo_pcs, BC_1	1.498689
pseudo_pcs, BC_2	−0.807297
pseudo_pcs, BC_3	3.507481
pseudo_pcs, BC_4	1.282007
pseudo_pcs, SO2_0	1.087772
pseudo_pcs, SO2_1	1.462824
pseudo_pcs, SO2_2	1.427859
pseudo_pcs, SO2_3	1.940259
pseudo_pcs, SO2_4	−1.740802

We fixed the methane, sulfur dioxide, and black carbon to SSP 245's year 2025 for our emulators. We performed Empirical Orthogonal Function (EOF) decomposition on Black Carbon (BC) and Sulfur Dioxide (SO_2). Methane is normalized where the maximum amount of methane is 0.8. Principal component time series are extracted to create five BC columns, each corresponding to one EOF mode's time series. The units for the principal component BC and SO_2 is Tg/year (teragrams per year).

Fixing the other greenhouse gases at a constant level allows us to better see the effect of CO_2, which is the most well know and prominent greenhouse gas. Setting the constant values at 2025 values is an intuitive choice because it is currently the year 2025, and we want to base our future predictions on top the current situation.

Given a final CO_2 concentration 2100, we interpolated the trajectory of atmospheric CO_2 concentration by linearly increasing/decreasing the carbon dioxide amount from 2015 to 2100, assuming equal step every year, to predict the yearly surface air temperatures. Linear interpolation was chosen because it does not assume overly complicated models and is applicable given any valid 2100 CO_2 concentration. The predicted series of temperatures was then used as an input for our sea level model.

2.2 Climate Model Emulators

Pattern Scaling

The pattern scaling model's performance is among the best relative to the other emulators in the ClimateBench, even though it is only based on a series of linear regressions. This model is limited by its inability to capture nonlinear relationships. If nonlinear relationships are present in different climate model runs, then we can expect the error for pattern scaling models to be a bit worse than what was observed before.

Gaussian Process Emulator

Climate systems are governed by complex, smooth, and highly nonlinear relationships, making Gaussian Process (GP) emulators well-suited for predicting future climate scenarios. Building on our previous research in "Utilizing Emulators to Explore the Climate Model Parameter Space," we chose to utilize the original GP model from ClimateBench as a foundation for our work. This approach leverages the flexibility and uncertainty quantification capabilities of GPs to improve climate predictions.

Random Forest Emulator

Random Forest is an ensemble method that combines multiple decision trees to improve predictive performance. While decision trees capture non-linear relationships well, they tend to overfit. Random Forest mitigates this by averaging predictions, reducing variance, and enhancing robustness. This makes it ideal for climate model emulation, where multiple target variables require separate models.

The features for our random forest model consist of the first five principal components of SO2 and BC, CO2, and CH4. `max_features` was changed from `'auto'` to `'sqrt'` to accommodate a different version of `rf_model` while achieving a similar result. The rest of the hyperparameters are tuned using random search of the training data without replacement. The final values for the hyperparameters are `'n_estimators': 250, 'min_samples_split': 5, 'min_samples_leaf': 7, 'max_depth': 5`.

CNN-LTSM Emulator

Neural networks excel at climate prediction because of their ability to model complex, non-linear relationships between atmospheric variables. Their deep architectures allow them to learn patterns from large-scale climate data, capturing intricate dependencies that traditional models may overlook. Their adaptability also allows them to generalize well across different climate scenarios. Making CNNs valuable for long-term forecasting and extreme weather prediction. We decided to use the original CNN-LTSM model from the ClimateBench.

2.3 Sea Level Rise Projection

Using the model described by Rahmstorf 2007, we then produce a linear fit for change in sea level height, regressed on temperature anomaly (temperature difference from a baseline). We take our temperature air surface variable from each of the emulator output files and then use it to train the model on predicted sea level rise in the NOR-ESM2 model for each SSP scenario.

Mathematically, the model equation is of the form:

$$\frac{dH}{dt} = a(T - T_0)$$

where $\frac{dH}{dt}$ is change in sea level height per year, a is a proportionality constant, and $T - T_0$ is temperature relative to a baseline. Finally to get the total sea level rise, we integrate the rate of sea level rise $\frac{dH}{dt}$, to get the total height at the final year of recorded temperature

$$H(t) = \int_{t_0}^{t} \frac{dH}{dt} dt.$$

Programmatically, this means using `np.cumsum()` to add up all the changes to obtain the year-over-year sea level rise. Additionally, depending on the source of the training sea level rise data, the data may need to be transformed by using `np.diff()` to turn total height into the rate of change of height.

Finally, as a simple sanity check, we compare visually and quantitatively the predicted sea level rise against both historical satellite data and other projections of sea level rise. We took NASA's projected sea level rise and modified it slightly by adding up the changes from 2015 to 2100, like the rest of our data (NASA 2021).

3 Results

3.1 Florida Sea Level Rise

As stated earlier, Florida is a state particularly susceptible to sea level rise due to its low-lying topography and extensive coastline. To visualize the rise in sea level and its impact on Florida, we made use of digital elevation models (DEM). DEMs are a representation of the bare ground topographic surface of the Earth, excluding trees, buildings, and any other surface objects. To generate DEMs, LIDAR data (essentially 3D scans of the Earth's surface) is taken and processed using algorithms, supplemented with other data sources, to construct the true elevation of the land surface.

For visualizing the topography of Florida corresponded to different sea level rise amounts, we chose the following coastal locations: Sanibel Island, Miami, Fort Myers Beach, the space in between Audubon and Merritt Island, and Everglades City. Given an emission scenario and the temperature projections, we took the median of predicted sea level rise in the year 2100 and determined how much of the land would be submerged. For this paper, we will focus on one scenario and use as input the cumulative amount of carbon dioxide in 2100 for SSP 245, which is about 4520 Gigatons.

For seeing sea level rise under this and other scenarios through interactive visualizations, visit our web application, SeeRise, or refer to the **Additional Figures** section.

3.2 Predicted Sea Level Rise

Expected Sea Level Rise

According to NASA projections, the expected cumulative rise in sea level under SSP 245 between 2015 and 2100 will be 536.4 mm ($\pm$ about 158 mm for the 66% confidence interval) (NASA 2021). This is about as large as a large pizza box. We modified the original data, which starts from 2011, for our range of years (2015-2100).

How did we calculate our sea level rise prediction?

Our predictions were calculated by keeping the methane, sulfur dioxide, and black carbon at 2025 levels under SSP 245 - see Table 1. Given the CO_2 concentration in 2100, we linearly interpolated the carbon dioxide concentration levels from 2015 to 2100, where it ended at 4520 Gigatons (for this particular paper).

In the Rahmstorf paper, a linear regression is trained on historical temperature and the difference between the predicted temperature and the average. This makes our pattern scaling emulator a fantastic one-to-one comparison when looking at predicted sea level rise. Our pattern scaling to sea level rise pipeline uses cumulative CO_2 to compute air surface temperature (TAS), which is then averaged and used to calculate the rate of sea level rise.

Comparisons for 2100

Table 2: Prediction Error Comparison

Emulator	Predicted (mm)	NASA Predicted - Emulator Predicted (mm)
Pattern Scaling	513.6	22.8
Gaussian Process	511.6	24.8
Random Forest	511.3	25.1
CNN-LSTM	417.0	119.4

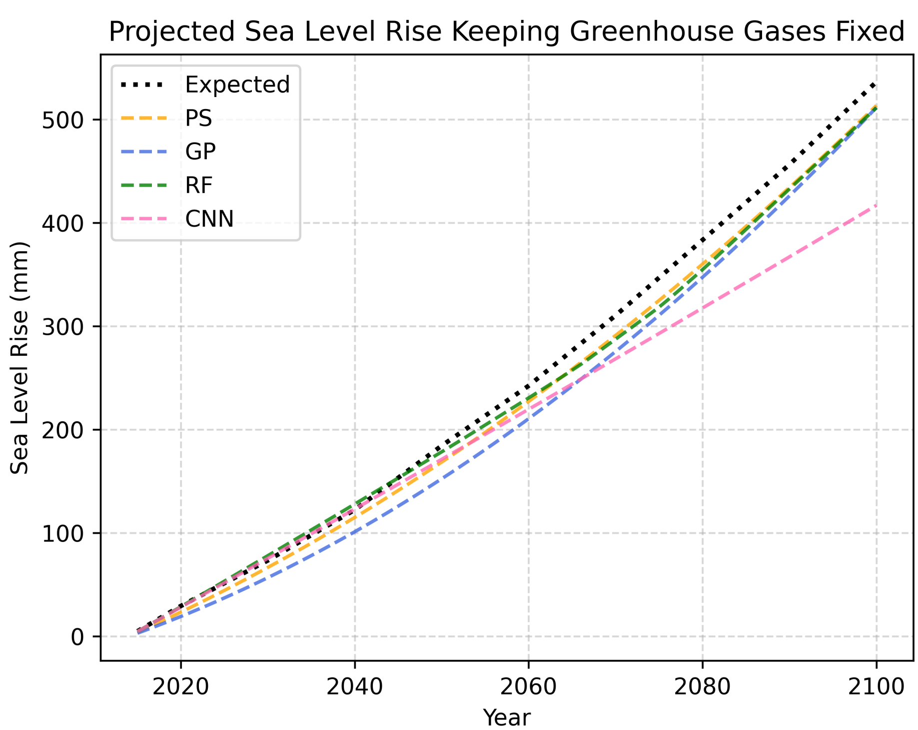

Figure 1: Sea Level Rise Predictions for emulators, calculated while keeping greenhouse gases at 2025 values, compared to NASA's expected sea level rise.

Validation Set Sea Level Rise Analysis

As we can see from Figure 1, the Pattern Scaling, Gaussian Process, and Random Forest emulators perform about equally well when compared to the expected sea level rise. They are under-predicting by about the size of a peanut (20 mm) or an inch (25 mm). The CNN-LTSM model performs the worst. The measurement is off from the expected value by about a standard playing card's length (120 mm).

The under-prediction exists likely because we decided to keep the other greenhouse gases constant at the values under SSP 245 from the year 2025. If the other greenhouse gases input were scaled appropriately alongside CO_2, the emulators would likely predict more accurately for TAS values, which in turn would lead to better performance of our sea level rise regression.

4 Discussion

Similar to past work, our sea level rise model using Rahmstorf's approach is prone to over-predicting when evaluated against the true expected projections. The model assumes a near-linear relationship between temperature and sea level rise rate, based on 20th-century observations. However, real-world ice sheet dynamics and ocean thermal expansion may not respond linearly to temperature changes, leading to over-estimations.

On the other hand, our model suffers from under-prediction when only changing CO_2 year to year and keeping the other greenhouse gases constant. Future work can be done on scaling all other greenhouse gases input appropriately, which would likely produce more accurate temperature values and sea level rise predictions.

5 Conclusion

Inspired by the pressing issue of sea level rise due to global warming, we worked on modeling projected sea level rise for the future up until 2100, and made use of climate model emulators as a first step for temperature inputs into the sea level projection model. The sea level rise projection model follows a semi-empirical differential equation approach from Rahmstorf, in which we fitted a linear model and obtained a rate of sea level rise, then integrated to get the total sea level rise. We also explored the impact of sea level rise on one particularly vulnerable region, Florida, by utilizing DEM data and visualizing the change in local topography and coastline following different amounts of sea level rise.

Overall, our work demonstrated the effectiveness of using machine learning models and differential equation approaches to achieve fairly sound results in predicting temperature and sea level rise. The interactive visualizations were created in the hope to make understanding impacts of sea level rise more intuitive and accessible for potential audiences and to raise further awareness of the issue of global warming and sea level rise.

References

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Appendices

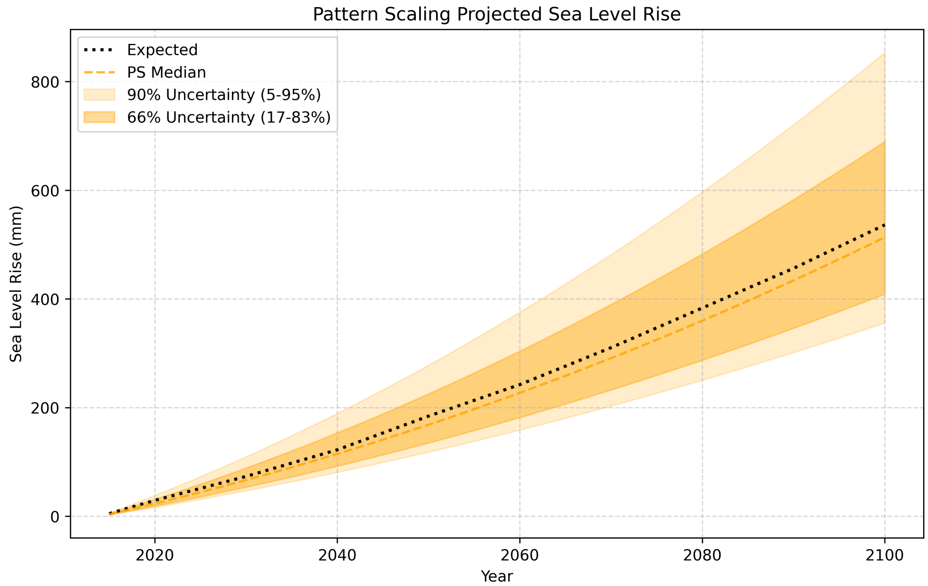

Figure A 1: Pattern Scaling sea level rise uncertainty compared to modified NASA's expected sea level rise (2015-2100).

A.1 Additional Figures

A.2 Contributions

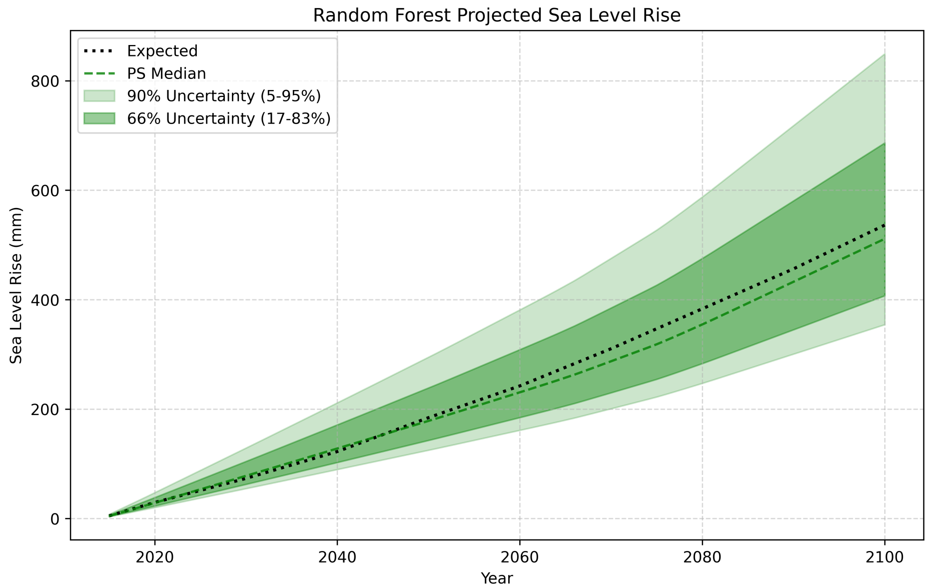

Figure A 2: Random Forest sea level rise uncertainty compared to modified NASA's expected sea level rise (2015-2100).

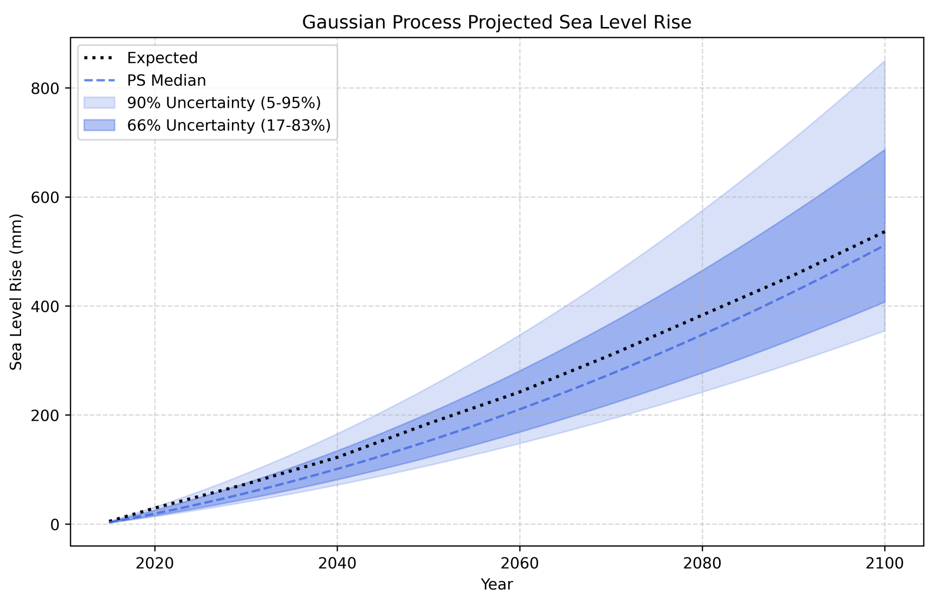

Figure A 3: Gaussian Process sea level rise uncertainty compared to modified NASA's expected sea level rise (2015-2100).

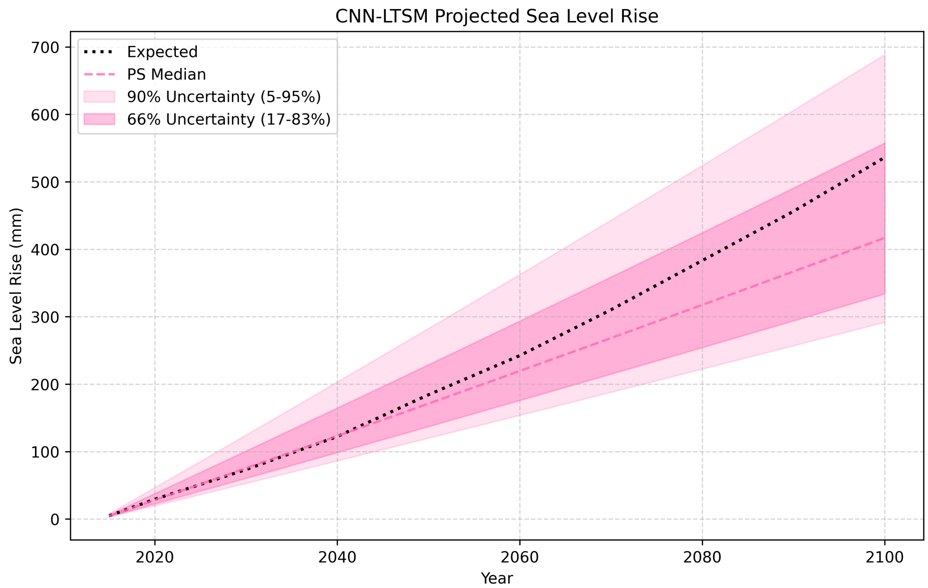

Figure A 4: CNN-LTSM sea level rise uncertainty compared to modified NASA's expected sea level rise (2015-2100).

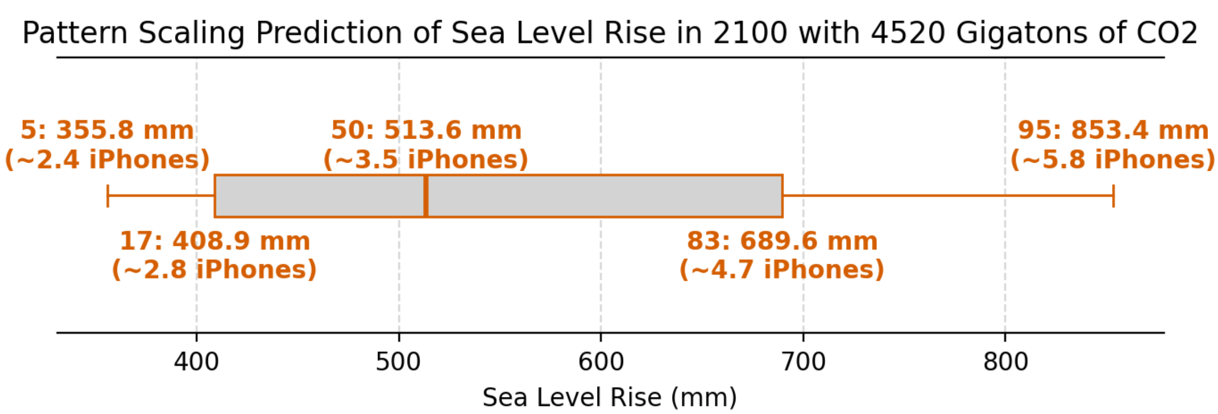

Figure A 5: Pattern Scaling's quartiles of sea level rise with 4520 Gigatons of carbon dioxide.

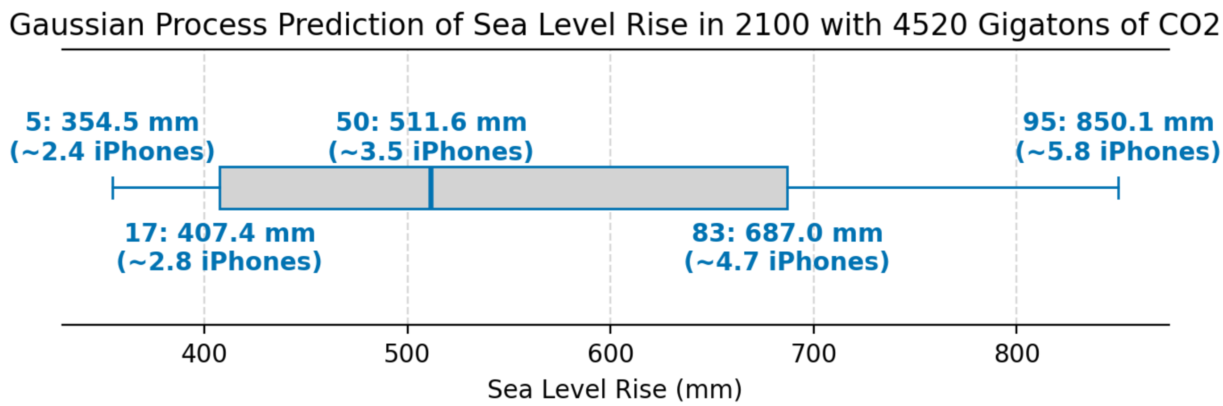

Figure A 6: Gaussian Process's quartiles of sea level rise with 4520 Gigatons of carbon dioxide.

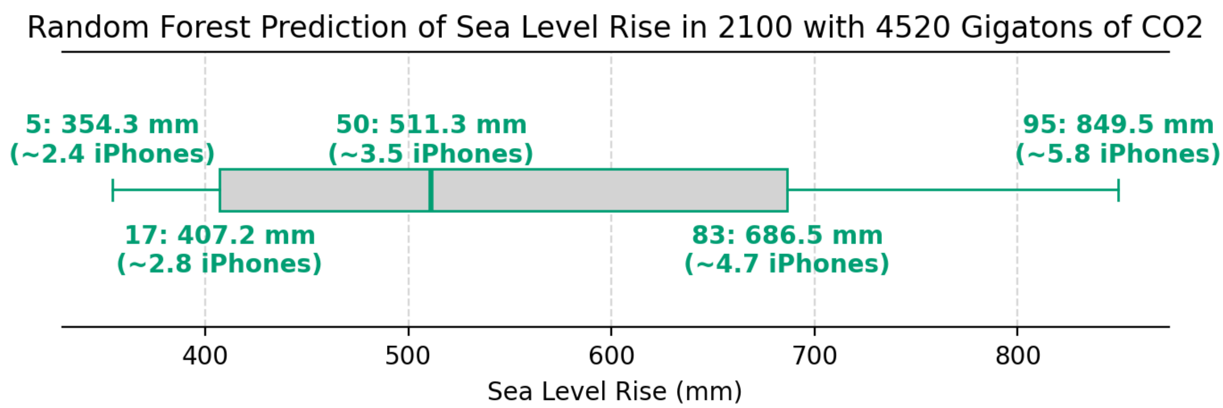

Figure A 7: Random Forest's quartiles of sea level rise with 4520 Gigatons of carbon dioxide.

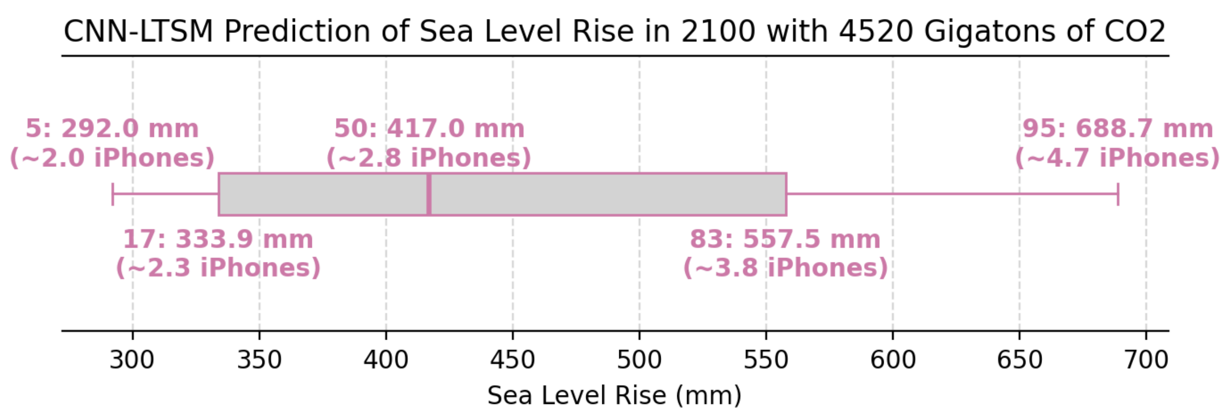

Figure A 8: CNN-LTSM's quartiles of sea level rise with 4520 Gigatons of carbon dioxide.

Zoe Ludena:

- Re-created the Gaussian Process Emulator from the ClimateBench (Watson-Parris 2022).
- Re-created the CNN-LTSM Emulator from the ClimateBench (Watson-Parris 2022).
  - Attempted finding hyper parameters, but was unsuccessful in producing something as good or better than original.
- Created the SeeRise website.
  - Created Figures (and content), Team (Zoe and Duncan's profiles), and App pages (embedded application).
- Created the SeeRise application.
  - Developed the front end, interactive components, and static figures.
  - Wrote commentary and explanations.
  - Added to DEM visualization.
  - Input datasets for Guassian Process and CNN-LTSM emulators for sea level rise.
- Added figures and writing to the poster.
- Edited and wrote Q2 Report.

Ylesia Wu:

- Re-created the Random Forest Emulator from the ClimateBench (Watson-Parris 2022).
  - Hyper parameter tuning : Improved the model compared to the original in terms of range of temperature predicted.
- Explored directly predicting sea level rise from TAS and year.
- Explored other interpolation methods for the trajectory of CO_2 concentrations.
- Created first draft of the poster.
  - Implemented poster requirements.
  - Organized content.
- Input datasets for Random Forest emulator for sea level rise for the SeeRise application
- Added personal bio to SeeRise website.
- Edited and wrote Q2 Report.

Eric Pham:

- Re-created the Pattern Scaling Emulator from the ClimateBench (Watson-Parris 2022).
- Developed code to create DEM visualizations on the SeeRise application.
- Implemented pipeline to re-create Rahmstorf's paper (Rahmstorf 2007).
- Modified NASA's expected value to match the years we used (2015-2100).
- Added personal bio to SeeRise website.
- Input datasets for Pattern Scaling emulator for sea level rise for the SeeRise application
- Added writing to the poster.
- Edited and wrote Q2 Report.